

Solution Requirements

Date	15-07-2026
Team ID	
Project Name	Rising Waters
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do).

Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users can securely access the Flood Prediction System through the web application. (Login can be added in future versions.)	High
2	Authorization Levels	Admin can manage the machine learning model and dataset, while users can enter environmental data and view flood predictions.	High
3	External Interfaces	The system uses a trained Machine Learning model (.pkl) and can be integrated with weather APIs or cloud services in future versions.	Medium
4	Transaction Processing	The system accepts user inputs, processes them using the ML model, and instantly generates flood prediction results.	High
5	Reporting	Displays prediction status, flood risk percentage, and graphical visualizations using charts.	High
6	Business Rules	Flood prediction is based on parameters such as rainfall, latitude, longitude, elevation, slope, distance, deaths, affected people, duration, and time.	High
7	Compliance to Laws or Regulations	User data is processed securely and only for prediction purposes while following basic data privacy principles.	Medium
8	Other	Provides a responsive web interface with easy navigation and real-time prediction results.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should quickly process user inputs and display prediction results.	Prediction generated in less than 3 seconds

2	Scalability	The application should support multiple users and allow future integration with cloud deployment.	Supports 500+ users with scope for future scaling
3	Security & Data Privacy	User input data should be processed securely without unauthorized access.	Secure processing with HTTPS support and protected application
4	Reliability & Availability	The application should provide consistent predictions with minimal downtime.	99% availability during normal operation
5	Usability & Accessibility	The dashboard should be simple, responsive, and user-friendly on desktops and laptops.	Easy-to-use interface with responsive design
6	Other (Maintainability & Portability)	The project should be easy to maintain, update, and deploy on different platforms such as local systems or cloud environments.	Modular Python and Flask code with GitHub version control